

# School of Economic, Political and Policy Sciences

## Geospatial Information Sciences (BS)

Geospatial Information Science (or GIScience) is the study of relationships between phenomena in space and time. In recent years, powerful new technologies and techniques have emerged that greatly improve our ability to acquire, archive, analyze, and communicate information regarding people, places, and other things on or near the Earth's surface. These same technologies and techniques allow us to combine this information into multi-tiered databases describing the physical, social, and other aspects of all or portions of the Earth. Such databases can then be analyzed in novel ways that take the data's explicit spatial (or locational) nature into account. The insights produced by analyzing these types of databases are revolutionizing many fields of science, government, and business. Currently, commonplace consumer products such as web-based mapping systems and GPS units that incorporate locational information are directly impacting the everyday lives of ordinary individuals.

Graduates of the Bachelors of Science in Geospatial Information Science program will understand the logical, mathematical, and technological foundations for compiling and analyzing spatial data. They will be skilled in solving geospatial problems, enabling them to move into professional roles handling the geospatial needs of typical corporate, government, and nonprofit organizations. The graduates will not only be skilled in the use of common GIScience software systems, but also will understand the underlying principles upon which software systems are based. This will allow them to transfer their knowledge from one software system to another, to expand the capabilities of these systems, and most importantly, to view geospatial problems as issues that can be solved by applying basic theories, techniques and methodologies.

## Mission and Objectives

The mission of the Bachelor of Science in Geospatial Information Sciences program is to provide students with a rigorous understanding of the fundamental theories and concepts underlying GIScience, as well as to provide them with extensive hands-on experience with contemporary GIScience hardware and software. The goal of the program is to give students a firm grasp of the theories, ideas, and techniques that underlay software and hardware systems for the compilation and analysis of spatially referenced data, and thus provide them with a foundation of knowledge and skill that transcends any individual piece of software or hardware. Graduates of this program will be able to successfully compete for professional positions within GIScience and related fields, and be admitted into the best graduate schools globally.

Students within the program will:

- Demonstrate their understanding of the underlying theories, ideas, concepts and techniques of

GIScience.

- Master contemporary computer hardware and software systems commonly employed in GIScience.
- Demonstrate problem solving skills that employ their understanding of theories, ideas and concepts as well as their mastery of GIScience software and hardware.

## Bachelor of Science in Geospatial Information Sciences

[Degree Requirements](#) (120 semester credit hours)

[View an Example of Degree Requirements by Semester](#)

### Faculty

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**Professors:** Yongwan Chun, Dohyeong Kim, David J. Lary, Fang Qiu, May Yuan

**Associate Professor:** Muhammad Rahman, Michael Tiefelsdorf

**Assistant-Professor:** Elías Cisneros, Alexander Michels

**Professor Emeriti:** Kurt J. Beron, Ronald Briggs, Daniel A. Griffith

**Lecturer:** Bryan Chastain

**I. Core Curriculum Requirements:** 42 semester credit hours

Curriculum Requirements can be fulfilled by other approved courses from institutions of higher education. The courses listed are recommended as the most efficient way to satisfy both Core Curriculum and Major Requirements at UT Dallas.

**Communication: 6 semester credit hours**

Select any 6 semester credit hours from [Communication Core](#) courses (see advisor)

**Mathematics: 3 semester credit hours**

[MATH 1325](#) Applied Calculus I

Or select any 3 semester credit hours from [Mathematics Core](#) courses (see advisor)

**Life and Physical Sciences: 6 semester credit hours**

Choose two courses from the following:

[ENVR 2302](#) The Global Environment<sup>1</sup>

or [GEOG 2302](#) The Global Environment<sup>1</sup>

or [GEOS 2302](#) The Global Environment<sup>1</sup>

[GEOS 1303](#) Physical Geology

[NATS 1311](#) The Universe, and Everything Else

[NATS 2333](#) Energy, Water, and the Environment

[PHYS 1301](#) College Physics I

Or select any 6 semester credit hours from [Life and Physical Sciences Core](#) courses (see advisor)

### **Language, Philosophy and Culture: 3 semester credit hours**

Select any 3 semester credit hours from [Language, Philosophy and Culture Core](#) courses (see advisor)

### **Creative Arts: 3 semester credit hours**

Select any 3 semester credit hours from [Creative Arts Core](#) courses (see advisor)

### **American History: 6 semester credit hours**

Select any 6 semester credit hours from [American History Core](#) courses (see advisor).

### **Government/Political Science: 6 semester credit hours**

Select any 6 semester credit hours from [Government/Political Science Core](#) courses (see advisor)

### **Social and Behavioral Sciences: 3 semester credit hours**

Choose one course from the following:

[CRIM 1301](#) Introduction to Criminal Justice

[CRIM 1307](#) Introduction to Crime and Criminology

[ECON 2301](#) Principles of Macroeconomics

[ECON 2302](#) Principles of Microeconomics

[GEOG 2303](#) World Regional Geography<sup>1</sup>

[SOC 1301](#) Introduction to Sociology

Or select any 3 semester credit hours from [Social and Behavioral Sciences Core](#) courses (see advisor)

**Component Area Option: 6 semester credit hours**

[EPPS 2301](#) Research Design in the Social and Policy Sciences

[EPPS 2302](#) Methods of Quantitative Analysis in the Social and Policy Sciences<sup>1</sup>

Or select any 6 semester credit hours from [Component Area Core](#) courses (see advisor)

II. Major Requirements: 40 semester credit hours

**Major Preparatory Courses: 7 semester credit hours beyond Core Curriculum**

[ENVR 2302](#) The Global Environment<sup>1</sup>

or [GEOG 2302](#) The Global Environment<sup>1</sup>

or [GEOS 2302](#) The Global Environment<sup>1</sup>

[EPPS 1110](#) Critical Issues in the Social Sciences

[EPPS 2302](#) Methods of Quantitative Analysis in the Social and Policy Sciences<sup>1</sup>

[GEOG 2303](#) World Regional Geography<sup>1</sup>

[GISC 2305](#) Spatial Thinking and Data Analytics

or [GEOS 2305](#) Spatial Thinking and Data Analytics

or [EPPS 2307](#) Digital Earth

or [GISC 2307](#) Digital Earth

or [GEOS 2307](#) Digital Earth

[GISC 2309](#) Principles of Geospatial Information Sciences

or [GEOG 2309](#) Principles of Geospatial Information Sciences

or [GEOS 2309](#) Principles of Geospatial Information Sciences

**Major Core Courses: 15 semester credit hours**

[GISC 2326](#) Computer Mapping and Geovisualization

[GISC 4381](#) Spatial Data Science

[GISC 4325](#) Introduction to Remote Sensing

or [GEOS 4325](#) Introduction to Remote Sensing

[GISC 4382](#) GIS Database and Modeling

[GISC 4386](#) Climate Change and Sustainable Solutions

or [ENVR 4386](#) Climate Change and Sustainable Solutions

**Major Related Course: 18 semester credit hours**

Choose 6 courses from the following:

[GISC 3336](#) Fundamentals of Sustainability

[GISC 3342](#) Artificial Intelligence for Environmental Social Science

[GISC 3352](#) Geospatial Intelligence Concepts and Applications

[GISC 4317](#) Python Programming for Social Science

[GISC 4328](#) Drone and Remote Sensing

[GISC 4363](#) Internet Mapping and Information Processing

[GISC 4368](#) GIS Analysis for Real Estate

[GISC 4384](#) Health and Environmental GIS

[GISC 4385](#) Advanced Applications in GIS

[GISC 4395](#) Remote Sensing Geophysics

or [GEOS 4395](#) Remote Sensing Geophysics

[GEOG 3331](#) Smart and Sustainable Cities

or [ENVR 3331](#) Smart and Sustainable Cities

[GEOG 3357](#) Spatial Dimensions of Health and Disease

[GEOG 3359](#) Human Migration and Mobility

[GEOG 3370](#) The Global Economy

[GEOG 3372](#) Population and Development

[GEOG 3377](#) Urban Planning and Policy

or [PA 3377](#) Urban Planning and Policy

[GEOG 4338](#) Hazard and Disaster Management

[MATH 2413](#) Differential Calculus

III. Elective Requirements: 38 semester credit hours

### **Prescribed Electives: 15 semester credit hours**

All students are required to take at least fifteen semester credit hours of prescribed upper-division elective courses.

### **Free Electives: 23 semester credit hours**

This requirement may be satisfied with lower- and upper-division courses from any field of study.

The plan must include sufficient upper-division courses to total 45 upper-division semester credit hours.

1. A Major requirement that also fulfills a Core Curriculum requirement. Semester credit hours are counted in Core Curriculum.

## **Minors**

Students must take a minimum of 18 semester credit hours for the minor, 12 of which must be upper-division semester credit hours. Students who take a minor will be expected to meet the normal prerequisites in courses making up the minor, and should maintain a minimum GPA of 2.000 on a 4.00 scale (C average). Semester credit hours may not be used to satisfy both the major and minor requirements; however, free elective semester credit hours or major preparatory classes may be used to satisfy the minor.

### **Minor in Geography**

18 semester credit hours

#### **Required Courses: 9 semester credit hours**

[GEOG 2302](#) The Global Environment

[GEOG 2303](#) World Regional Geography

[GEOG 2309](#) Principles of Geospatial Information Sciences

## Upper-Division Courses: 9 semester credit hours

Any upper-division Geography (GEOG) or Geographic Information Sciences (GISC) courses, excluding [GEOG 4V97](#).

## Minor in Geospatial Information Sciences (GIS)

18 semester credit hours

### Required Courses: 9 semester credit hours

[GISC 2305](#) Spatial Thinking and Data Analytics

or [GISC 2307](#) Digital Earth

[GISC 2309](#) Principles of Geospatial Information Sciences

or [GEOG 2309](#) Principles of Geospatial Information Sciences

[GISC 4325](#) Introduction to Remote Sensing

### Upper-Division Courses: 9 semester credit hours

Any upper-division Geography (GEOG) or Geographic Information Sciences (GISC) courses, excluding [GEOG 4V97](#).

## Certificates

### Certificate in Geospatial Intelligence

15 semester credit hours

The Geospatial Intelligence (GEOINT) certificate program focuses on the technical expertise, rapid critical thinking, and global awareness necessary for supporting national security and other intelligence activities. The curriculum adheres to the rigorous requirements set by the United States Geospatial Intelligence Foundation (USGIF).

The coursework is delivered using the state-of-the-art facilities within the Geospatial Information Sciences program, which is housed in the School of Economic, Political and Policy Sciences. The certificate requires 15 undergraduate semester credit hours (five courses), detailed below. Importantly, all credits earned for this certificate can be applied toward the [Bachelor of Science in Geospatial Information Sciences](#). Other UT Dallas majors may need to meet the prerequisite requirements to complete the certificate program.

Since the program is accredited by the USGIF, students have the unique opportunity to earn the official USGIF GEOINT Certificate in addition to the UT Dallas GEOINT Certificate. Achieving this dual recognition requires students to complete a total of 18 semester credit hours (an additional 3 semester credit hours) and to select [GISC 4V97](#) as their elective. Furthermore, students must compile and submit a portfolio of their geospatial work--a requirement often fulfilled by the final project from [GISC 4V97](#)--to validate their skills to the USGIF. To qualify for the USGIF certificate, students must, at a minimum, have a cumulative average of "B" for all of the courses for the certificate, and get a "C" or better in each one.

Required Coursework (15 semester credit hours)

**Three required courses (9 semester credit hours):**

[GISC 4325](#) Introduction to Remote Sensing

or [GISC 4328](#) Drone and Remote Sensing

[GISC 4381](#) Spatial Data Science

[GISC 4317](#) Python Programming for Social Science

**Two elective courses chosen from the following(6 semester credit hours):**

[GISC 3342](#) Artificial Intelligence for Environmental Social Science

[GISC 4325](#) Introduction to Remote Sensing

or [GISC 4328](#) Drone and Remote Sensing

[GISC 4363](#) Internet Mapping and Information Processing

[GISC 4382](#) GIS Database and Modeling

[GISC 4V97](#) Independent Study in Geospatial Information Science

Or other courses as approved by the Director of the certificate program

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